

Module 13 — Troubleshooting & Reference

Supplementary to the sPHENIX Standalone Course

1. General strategy

When a component stops working, isolate the layer: is the tool configured correctly, is it running, is the physics/output sensible? Work top-down and never skip steps — the fastest path to the bug is usually from the outside in.

2. Environment / PATH problems

Symptom	Diagnosis
command not found: root	ROOT not sourced — <code>source \$HOME/opt/root/bin/thisroot.sh</code>
undefined reference to HepMC	Link order — library after object files in g++ command
cannot open shared object file	<code>LD_LIBRARY_PATH</code> missing
Pythia settings silently ignored	Flag type — Pythia logs unknown flags to stderr

3. Geant4 runtime errors

Error message	Meaning / fix
G4LEVELGAMMADATA not set	Source <code>geant4.sh</code> before running
Cannot open OpenGL viewer	Run headless or install X11 + OpenGL
EM process has no model for physics list 0	Physics list too narrow — use FTFP_BERT
Overlap warning: volume X Daughters bigger than mother	Daughters bigger than mother — shrink daughter
G4PhysicsTable has not been initialized	Primary generator fired before /run/initialize

4. HIJING problems

Symptom	Cause
Random crashes	Rebuild with <code>-fno-automatic</code>
Arithmetic exception	Impact parameter 0 with small nuclei — set BMIN > 0.01
Events with N_ATT = 0	Projectile didn't hit target — increase BMAX
Non-reproducible results with fixed seed	HIPNDSeed not set; set it explicitly in driver

5. Pythia gotchas

Symptom	Fix
Events take forever to generate	ptFlatMin too low; use 2-5 GeV for min-bias, 20+ for jets
Zero events after setting a process	Process-space cuts conflict with process threshold
Charmonium on, but no J/ψ decays	Decays disabled — check 443:onMode
HepMC file has zero particles	Writer not flushed — wrap writer loop in scope

6. ROOT output issues

Symptom	Fix
TTree is empty	Forgot <code>tree.Fill()</code> or <code>tree.Write()</code>
<code>Draw()</code> opens ROOT in batch mode	ROOT in batch mode; use <code>-b</code> only intentionally
Histogram values look like $\pm 1e300$	Uninitialised memory — always default-initialize branch vars
Gaussian fit fails: minimiser broke	Provide <code>SetParameters()</code> near best guess

7. Debugging workflow

When something's wrong, always: (1) reduce to minimal test case, (2) check logs completely, (3) verify by printing, then (4) use a debugger.

1. Verbosity

/tracking/verbose 2

/process/verbose 1

2. Debugger

`gdb --args ./exampleB4a -m run.mac`

`(gdb) run`

`(gdb) bt`

3. Valgrind memory check

`valgrind --tool=memcheck --leak-check=full ./exampleB4a -m run.mac`

8. Quick-reference cheat sheets

Pythia8 essentials

```
Beams:ecm = 200.  
HardQCD:all = on  
PhaseSpace:pTHatMin = 20.  
Tune:pp = 14  
Random:setSeed = on  
Random:seed = 42  
Next:numberShowEvent = 1
```

Geant4 macro essentials

```
/run/initialize  
/gun/particle e-  
/gun/energy 10 GeV  
/gun/position 0 0 0  
/gun/direction 0 0 1  
/run/beamOn 1000
```

ROOT CLI basics

```
root -l file.root  
root [0] tree->Print();  
root [1] tree->Draw("pt>>h(100,0,20)", "pt>0.5");  
root [2] h->Fit("expo");  
root [3] .q
```

9. Physics sanity checks

Variable	Expected at RHIC (200 GeV)
Inelastic pp σ	~ 42 mb
$\langle dN_{\text{ch}}/d\eta \rangle$ min-bias pp	~ 2.4 at $\eta=0$
$\langle p_T \rangle$ charged pions	~ 0.5 GeV/c
J/ψ cross section $\times$ BR(ee)	~ 8 nb
$dN_{\text{ch}}/d\eta$, Au+Au 0-5%	~ 680

v_2 of charged π at p_T=2 GeV	GM 20-30%
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10. When all else fails

- Take a break. Seriously — most simulation bugs reveal themselves after 30 minutes away.
- Re-read the output of configure / CMake carefully — warnings there often predict runtime failures.
- Run the example you didn't write. If B4a works and yours doesn't, the bug is in your code.
- Compare two git commits and bisect.
- Search the Geant4 / sPHENIX mailing-list archives — someone has hit this before.